Testing Predictors & Evaluating Linear Models

Revision sheet — Data Analysis for Psychology in R 2 (DAPR2), Week 3 · University of Edinburgh 2026–27

Learning objectives (= what can be examined)

- Interpret **significance tests for β coefficients**.
- Calculate and interpret **R^2** and **adjusted R^2** as measures of model quality.
- Locate the significance of individual predictors and overall model fit in R's `lm` output.

1. Recap and the three parts of model evaluation

Running model: $\text{Score}_i = \beta_0 + \beta_1 \cdot \text{Hours}_i + \beta_2 \cdot \text{Motivation}_i + \varepsilon_i$, fitted with `lm(score ~ hours + motivation, data = test_study2)` (n = 150).

Evaluating a model means evaluating…	When
1. The **individual coefficients** (is each predictor significant?)	This week
2. The **overall model quality** (how much variance does it explain?)	This week
3. The **model assumptions**	Semester 1, Week 8

All three must be considered before accepting a set of results.

2. From research question to statistical hypothesis

"Is study time a *useful predictor* of test score?" is vague: we must turn it into a **testable statistical hypothesis**.

Steps in hypothesis testing

Research question → statistical hypothesis → define the **null** → calculate the **estimate** of the effect → calculate a **test statistic** → evaluate the test statistic **against the null**.

- A good **research question** defines the constructs, the association tested, its **direction** and the target population. E.g. *Does increased study time improve test scores in school-age children?*
- **H_0** is **precise**: the effect of interest equals zero. **H_1** is **not specific**: the effect is not zero (no claim about its actual size).
- We can **only ever test the null**. Choose α (usually .05) and compute the p-value.
- **p-value** = probability of observing our test statistic **or a more extreme one, assuming H_0 is true**.
- $p < \alpha$ → **reject H_0**; $p \geq \alpha$ → **fail to reject H_0** (never "accept H_0").

3. Defining the null for a slope

If x and y are unrelated, a change in x produces **no change** in the predicted y, so the slope is 0. Hence:

$$\mathbf{H_0: \beta_1 = 0} \qquad \mathbf{H_1: \beta_1 \neq 0}$$

4. The t-test for a β coefficient

t = b / SE(b) = estimated coefficient / its standard error

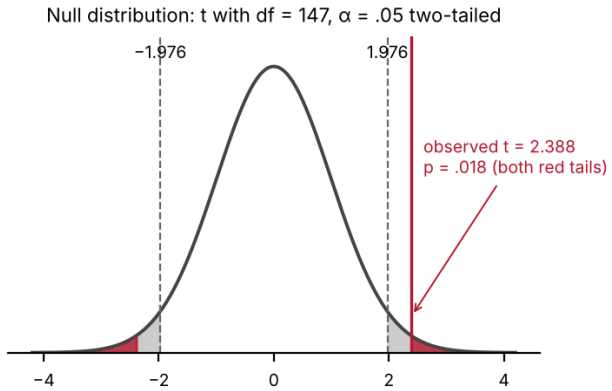
- The **standard error** describes the spread of the sampling distribution of the estimate. **Smaller SE = more precise estimate** (good).
- Under H_0, t follows a **t-distribution with n − k − 1 degrees of freedom**: n = sample size, k = number of predictors (βs excluding the intercept), −1 for the intercept.

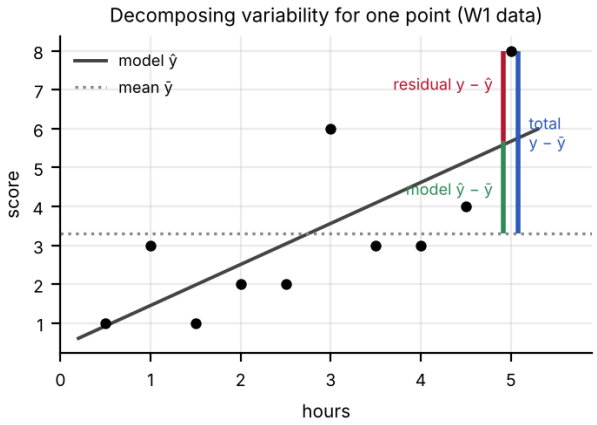
```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.86679    0.65473  10.488   <2e-16 ***
hours        1.37570    0.07989  17.220   <2e-16 ***
motivation   0.91634    0.38376   2.388   0.0182 *
---
Residual standard error: 4.386 on 147 degrees of freedom
Multiple R-squared: 0.6696,     Adjusted R-squared: 0.6651
F-statistic: 148.9 on 2 and 147 DF, p-value: < 2.2e-16
```

Worked example: is motivation a significant predictor?

Motivation (β_2): t = 0.9163 / 0.3838 = **2.388**. df = 150 − 2 − 1 = **147** (matches "on 147 degrees of freedom").
Critical values for α = .05 two-tailed: `qt(c(0.025, 0.975), 147)` → **±1.976**.
|2.388| > 1.976 → in the rejection region. p = .018 < .05 → **reject H_0**: the association between motivation and score (controlling for hours) is significantly different from zero.
Hours: t = 1.3757 / 0.07989 = 17.22, p < .001 → also significant.

- **Critical values (t*)** mark the boundary of significance: if H_0 is true, the probability that t falls beyond them is α. For a **two-tailed** test, α is split: 2.5% in each tail.
- **p-value** = area beyond the observed t in one tail, **× 2** for a two-tailed test.
- α = the risk of **incorrectly rejecting a true null** (Type I error).





Left: t = 2.388 lies beyond the critical values; the red tails give p = .018. Right: for any point, (y − ȳ) = (ŷ − ȳ) + (y − ŷ), i.e. total = model + residual.

5. Model quality: the coefficient of determination R^2

Variability in y has two sources: variability **the model can explain** (because x affects y) and random variability **it cannot explain**. A model is judged by the **proportion of variability it explains**.

Sum of squares	Formula	In words
$\mathbf{SS_{Total}}$ total variability	$\Sigma (y_i - \bar{y})^2$	Observed value minus the **mean** of y, squared, summed.
$\mathbf{SS_{Residual}}$ not explained	$\Sigma (y_i - \hat{y}_i)^2$	Observed value minus the **model's prediction**, squared, summed.
$\mathbf{SS_{Model}}$ explained	$\Sigma (\hat{y}_i - \bar{y})^2$	Model's prediction minus the **mean** of y, squared, summed.

$$\mathbf{SS_{Total} = SS_{Model} + SS_{Residual}}$$

$$\mathbf{R^2 = SS_{Model} / SS_{Total} = 1 - SS_{Residual} / SS_{Total}}$$

- R^2 = the **proportion of variability in the outcome accounted for by the predictors** (between 0 and 1).
- The more variance accounted for, the **better the model fit**.
- It shows how much predicting y is **improved** by using the linear relation with x, compared with not using x (i.e. just predicting $\bar{y}$ for everyone).

Worked example

SS_{Total} = 8556.06, $SS_{Residual}$ = 2826.83, SS_{Model} = 5729.23 (check: 5729.23 + 2826.83 = 8556.06).
R^2 = 5729.23 / 8556.06 = **0.6696** = R's "**Multiple R-squared**".
→ *67% of the variation in test scores is accounted for by hours of revision and student motivation.*

6. Adjusted R^2

With two or more predictors, R^2 **overestimates** the population value:

- Through random sampling fluctuation, even when $R^2 = 0$ in the population, the sample $R^2 \neq 0$.
- Fluctuations are larger in **smaller samples**.
- **More predictors** = more opportunities to add positive fluctuation (R^2 never decreases when you add a predictor).

$$\mathbf{adjusted\ R^2 = 1 - (1 - R^2) \cdot (n - 1) / (n - k - 1)}$$

It adjusts for **sample size n** and **number of predictors k**, and is the more **conservative** estimate.

Worked example

1 − (1 − 0.6696) × 149 / 147 = **0.6651** = R's "**Adjusted R-squared**".
→ *Based on adjusted R^2, hours studying and motivation explain 66.5% of the variance in test scores.*
R^2 (0.670) and adjusted R^2 (0.665) are close because **n is large and k is small**. With more predictors or a smaller sample, the gap widens and adjusted R^2 is the more accurate figure.

7. Where everything is in the lm output

```
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What you want	Where to find it	Example value
Estimate of each β	**Estimate** column	motivation: 0.916
Precision of the estimate	**Std. Error** column	0.384
Test statistic for H_0: β = 0	**t value** = Estimate / Std. Error	2.388
Significance of a predictor	**Pr(>|t|)** (two-tailed p) and stars	0.0182 *
Degrees of freedom (n − k − 1)	"Residual standard error: … on **147** degrees of freedom"	147
Overall variance explained	**Multiple R-squared**	0.6696
Conservative variance explained	**Adjusted R-squared**	0.6651
Overall model test	**F-statistic** line (covered next week)	148.9 on 2 and 147 DF

8. Appendix: what makes SE small? (formula not to memorise, logic to understand)

$$SE(b_j) = \sqrt{[\ (SS_{Residual} / (n - k - 1)) / (\ \Sigma(x_{ij} - \bar{x}_j)^2 \cdot (1 - R^2_{x_i})\ )\ ]}$$

SE is **smaller** (more precise, easier to reach significance) when...	SE is **larger** when...
• residual variance ($SS_{Residual}$) is small • the sample size n is large	• the number of predictors k is large • the predictor is strongly correlated with the other predictors ($R^2_{x_j}$ high; $R^2_{x_j}$ = 0 with one predictor)

Quick self-test (answers included)

For the Week 1 model (n = 10, one predictor): t = 1.0545 / 0.3581 = **2.945**; df = 10 − 1 − 1 = **8**; critical t = ±2.306; p = .019 → slope significant.
Adjusted R^2 = 1 − (1 − 0.5201) × 9/8 = **0.460**: much lower than R^2 = 0.520, because **n is small**.

Must remember

- Evaluate a model on 3 fronts: coefficients, overall quality, assumptions.
- H_0: β = 0. Test with **t = b / SE(b)**, df = **n − k − 1**; reject if p < α (equivalently |t| > t*).
- Significance is more likely with **small residuals** and a **large sample**.
- **$R^2 = SS_{Model}/SS_{Total} = 1 - SS_{Residual}/SS_{Total}$**: proportion of variance explained.
- With multiple predictors, report **adjusted R^2** (more conservative): $1 - (1 - R^2)(n - 1)/(n - k - 1)$.